

used for real world, high-performance computing applications that include:

- Fault-tolerant qubits
- Solar energy prediction
- AI for algorithm design
- Divisive clustering
- Molecular generation
- Circuit synthesis

Working with CUDA-Q

To install CUDA-Q, use:

```
$ pip install cudaq
```

In addition to a local laptop, CUDA-Q can be used on Google Colab.

Here's an example of sentiment analysis using a pre-trained model and CUDA-Q package:

```
# Import Packages
import cudaq
from transformers import pipeline

# Pretrained model of sentiment for
# integration with quantum
sentimentanalysis = pipeline(
    "sentiment-analysis",
    model="distilbert/distilbert-base-uncased-finetuned-sst-2-english"
)

mytext = "I really enjoyed this course!"

myresult = sentimentanalysis(mytext)[0]

print("Text:", mytext)
print("Sentiment:", myresult)

# Conversion of sentiment confidence to a
# quantum rotation
myscore = myresult["score"]

# CUDA-Q circuit
currentkernel = cudaq.make_kernel()
q = currentkernel.qalloc()

currentkernel.ry(score * 3.14159, q)
currentkernel.mz(q)

# Quantum simulation
```

```
model.safetensors: reconstructing file: 100% 268MB / 268MB, 25.0MB/s
model.safetensors: downloading bytes: 250MB, 22.8MB/s
Loading weights: 100% 104/104 [00:00<00:00, 3004.10it/s]
tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 2.76kB/s]
vocab.txt: 100% 232k/232k [00:00<00:00, 6.33MB/s]
Text: I really enjoyed this course!
Sentiment: {'label': 'POSITIVE', 'score': 0.9998770952224731}
Quantum result: { 1:100 }
```

Figure 3: Output from Google Colab using CUDA-Q on sentiment analysis with pre-trained model



Figure 4: CUDA-Q Academic learning pathways

```
analysiscounts = cudaq.sample(kernel,
shots_count=100)

print("Quantum result:", analysiscounts)
```

CUDA-Q Academic

Nvidia's CUDA-Q Academic (<https://github.com/NVIDIA/cuda-q-academic>) program is a collection of open source study materials and interactive resources for academics and researchers, which can be used freely

for their research work and academic courseware. The portal offers easy to use and understand educational resources for academicians as well as researchers, with the integration of learning pathways.

Nvidia has given an excellent push to the development of quantum computing with its CUDA-Q platforms. By open sourcing them, the tech giant has made these platforms and programs highly available. **END** 🐧

References

- NVIDIA CUDA-Q, <https://developer.nvidia.com/cuda-q>
- NVIDIA CUDA-Q Quantum Documentation, <https://nvidia.github.io/cuda-quantum/latest/index.html>
- NVIDIA CUDA-Q Academic, <https://github.com/NVIDIA/cuda-q-academic>
- CUDA-Q Academic Learning Pathways, <https://nvidia.github.io/cuda-q-academic/learningpath.html>

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